



Flow Injection Analysis-Tandem Mass Spectrometry Method for The Analysis of Plasmalogenethanolamines and Phosphotidylethanolamines In Serum

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Summary (or Abstract)

Please describe the analysis performed (Should not exceed 200 words)

Method

Serum samples were prepared using liquid-liquid extraction for the analysis of 10 plasmalogens (PlsEtns) and 10 phosphoethanolamines (PtdEtns). 20 μ L of serum was extracted by adding 20 μ L of water and 600 μ L of ethyl acetate containing internal standards (ISs) D₉-PlsEtn 18:0/18:1 and D₃₁-PtdEtn 16:0/18:1, followed by a thorough mixing for 30 min at 1500 rpm on an Eppendorf Mixmate shaker. Samples were then centrifuged for 5 mins at 2000 rcf, and the organic supernatants transferred into a HPLC vial for analysis.

Samples were analyzed by flow injection tandem mass spectrometry (FIA-MS/MS) on a Sciex 4000 API triple-quadrupole mass spectrometer equipped with an atmosphere chemical ionization (APCI) source coupled with an Agilent 1200 HPLC system consisting of degasser, pump, and autosampler. 100 μ L of extract at room temperature was injected into the ion source using a mobile phase consisting of 80% ethyl acetate, 15% methanol, and 5% water at a flow rate of 0.45 mL/min. Calibration curves were constructed by plotting the ratio of serially-diluted analyte peak areas/IS peak areas versus theoretical concentration using a linear regression and $1/x$ weighting. PlsEtn concentrations were extracted off the PlsEtn 18:0/22:6 curve, while PtdEtns off the PtdEtn 16:0/22:6 curve. Acceptance criteria were $R^2 > 0.98$, less than 20% deviation for CC_01 (LLOQ) and 15% deviation for other calibrators, less than 15% deviation for at least 67% of QC samples, and at least 50% of QCs less than 15% deviation from their theoretical concentrations.

